

Lecture 21

Download docker

Download Docker Desktop Installer dari website resmi Docker, klik *Download for Windows*

Jalankan installer `Docker Desktop Installer.exe` (klik 2x)

Konfigurasi — centang opsi *Use WSL 2*, lalu klik OK

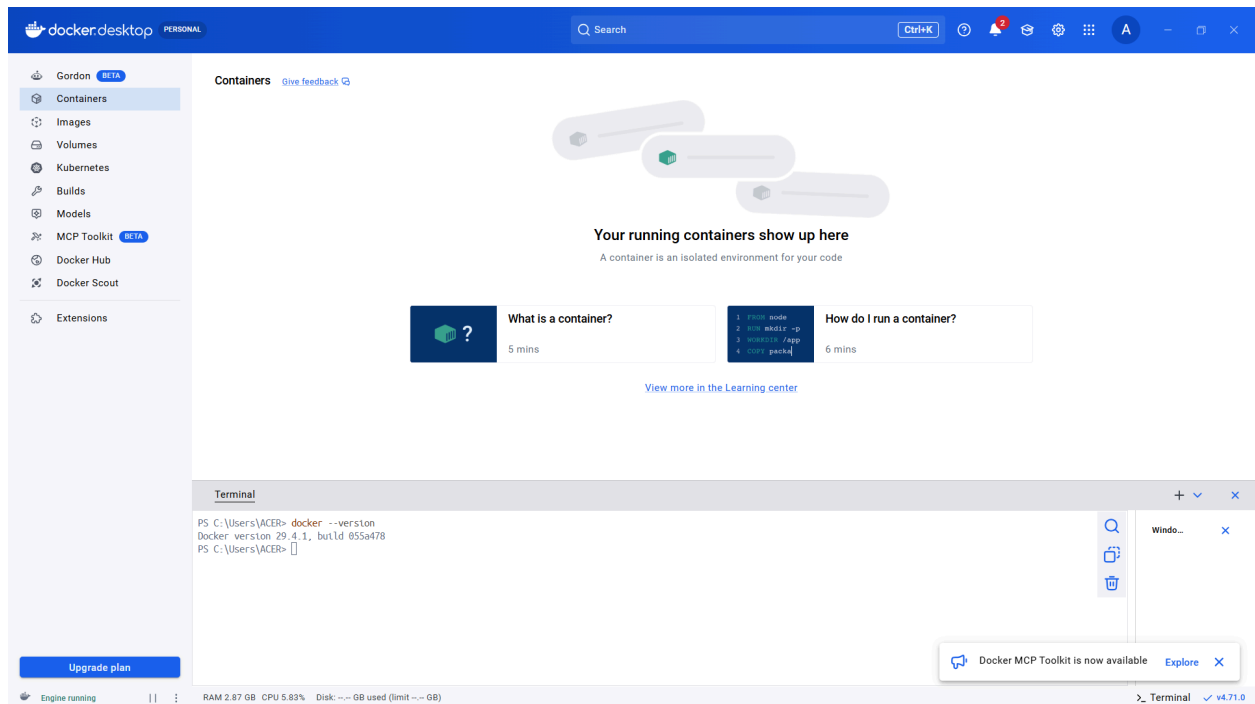
Tunggu proses instalasi selesai, lalu klik *Close and restart*

Terima agreement — setelah restart, centang *I accept the terms* dan klik Accept

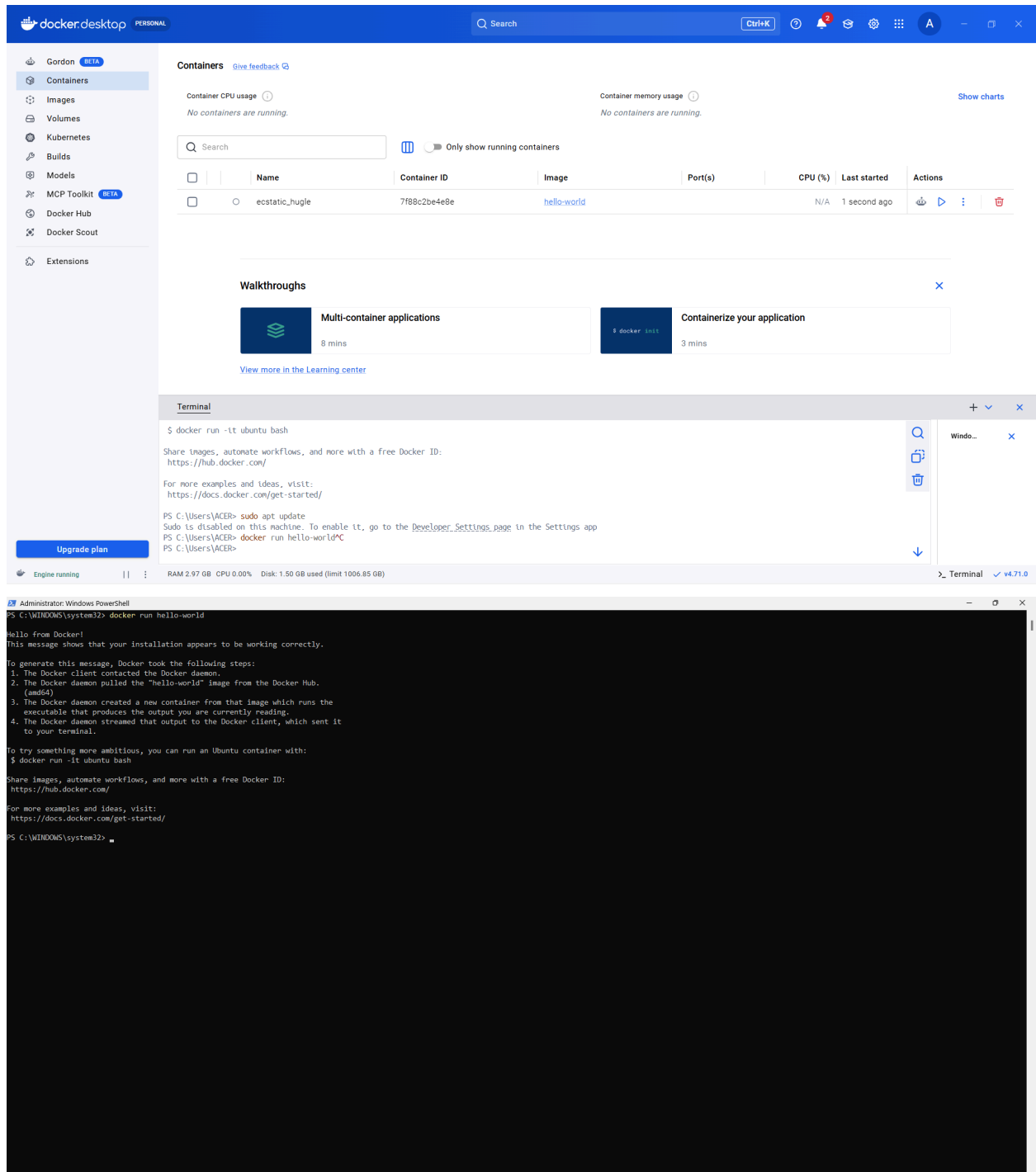
Install WSL 2 — klik link di jendela Docker Desktop untuk download, jalankan `ws1_update_x64.exe`, klik Next, tunggu selesai, lalu klik Finish dan restart

Verifikasi — buka CMD/PowerShell, jalankan `docker --version`

Install wsl



Hello world



Lecture 22

Command Top berfungsi untuk menampilkan kinerja sistem secara real time seperti CPU, Memory, Serta proses yang sedang berjalan.

Ketikkan Command dibawah ini Terminal:

> top

> Ctrl+c (exit)

```
root@nayakvpc: /home/nayak
top - 15:18:47 up 10 min, 2 users, load average: 0.01, 0.10, 0.07
Tasks: 109 total, 1 running, 108 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.3 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1963.6 total, 968.8 free, 527.4 used, 467.3 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 1278.9 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 780 mysql     20   0 1784296 395132 36256 S   1.0  19.7   0:05.86 mysqld
1469 root       20   0  10944    3688   3072 R   0.3   0.2   0:00.01 top
    1 root       20   0  102100  13108   8476 S   0.0   0.7   0:02.81 systemd
    2 root       20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 rcu_gp
    4 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
    5 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
    6 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 netns
    7 root       20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0-cgroup_release
    8 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
    9 root       20   0      0      0      0 I   0.0   0.0   0:00.07 kworker/u4:0-flush-252:0
   10 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
   11 root       20   0      0      0      0 S   0.0   0.0   0:00.00 rcu_tasks_rude_
   12 root       20   0      0      0      0 S   0.0   0.0   0:00.00 rcu_tasks_trace
   13 root       20   0      0      0      0 S   0.0   0.0   0:00.03 ksoftirqd/0
   14 root       20   0      0      0      0 I   0.0   0.0   0:00.06 rcu_sched
   15 root       rt   0      0      0      0 S   0.0   0.0   0:00.00 migration/0
   16 root      -51   0      0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
   18 root       20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
   19 root       20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
   20 root      -51   0      0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
   21 root       rt   0      0      0      0 S   0.0   0.0   0:00.09 migration/1
   22 root       20   0      0      0      0 S   0.0   0.0   0:00.03 ksoftirqd/1
   24 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
   25 root       20   0      0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
   26 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
   27 root       20   0      0      0      0 S   0.0   0.0   0:00.00 kauditd
   28 root       20   0      0      0      0 S   0.0   0.0   0:00.00 khungtaskd
   29 root       20   0      0      0      0 S   0.0   0.0   0:00.00 oom_reaper
   30 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 writeback
   31 root       20   0      0      0      0 S   0.0   0.0   0:00.02 kcompactd0
   32 root       25   5      0      0      0 S   0.0   0.0   0:00.00 ksmd
   33 root       39  19      0      0      0 S   0.0   0.0   0:00.00 khugepaged
   38 root       20   0      0      0      0 I   0.0   0.0   0:00.06 kworker/1:1-events
   80 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kintegrityd
   81 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kblockd
   82 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 blkcg_punt_bio
   83 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 tpm_dev_wq
   84 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 ata_sff
   85 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 md
   86 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 edac-poller
   87 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 devfreq_wq
   88 root      -51   0      0      0      0 S   0.0   0.0   0:00.00 watchdogd
   89 root       20   0      0      0      0 I   0.0   0.0   0:00.36 kworker/u4:1-flush-252:0
   90 root        0 -20      0      0      0 I   0.0   0.0   0:00.02 kworker/0:1H-kblockd
   92 root       20   0      0      0      0 S   0.0   0.0   0:00.00 kswapd0
   93 root       20   0      0      0      0 S   0.0   0.0   0:00.00 ecryptfs-kthrea
   95 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kthrotld
   96 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 acpi_thermal_pm
   98 root       20   0      0      0      0 S   0.0   0.0   0:00.00 scsi_eh_0
   99 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 scsi_tmf_0
  100 root       20   0      0      0      0 S   0.0   0.0   0:00.00 scsi_eh_1
  101 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 scsi_tmf_1
  103 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 vfio-irqfd-clea
```

Command Htop merupakan perintah untuk memonitoring proses yang sedang berjalan pada sistem melalui terminal linux. Sama seperti perintah Top, Htop berfungsi untuk memantau penggunaan memori, total swap yang digunakan, total CPU yang sedang berjalan, memantau aplikasi apa saja yang sedang berjalan, dan

seberapa lama komputer sudah berjalan.
Ketikkan Command dibawah ini Terminal:

> htop

> Ctrl+c OR F10 (exit)

```
root@nayakvpc: /home/nayak

0[          0.0%] Tasks: 38, 59 thr; 1 running
1[          0.0%] Load average: 0.00 0.09 0.06
Mem[|||||] 531M/1.92G Uptime: 00:11:01
Swp[          0K/0K]

  PID USER      PRI  NI  VIRT   RES   SHR  S  CPU% MEM%   TIME+  Command
 780 mysql      20    0 1742M  385M  36256 S   1.3 19.7  0:06.09 /usr/sbin/mysqld
 930 mysql      20    0 1742M  385M  36256 S   1.3 19.7  0:03.02 /usr/sbin/mysqld
 889 mysql      20    0 1742M  385M  36256 S   0.7 19.7  0:00.04 /usr/sbin/mysqld
 927 mysql      20    0 1742M  385M  36256 S   0.7 19.7  0:00.25 /usr/sbin/mysqld
1470 root         20    0   8756   4192   3272 R   0.7  0.2  0:00.02 htop
    1 root         20    0    99M  13108   8476 S   0.0  0.7  0:02.81 /sbin/init
 353 root        19   -1 70124 26484 25416 S   0.0  1.3  0:00.32 /lib/systemd/systemd-journald
 384 root         20    0   282M  27100   9072 S   0.0  1.3  0:00.10 /sbin/multipathd -d -s
 389 root         20    0   282M  27100   9072 S   0.0  1.3  0:00.00 /sbin/multipathd -d -s
 390 root         20    0 23004   6476   4744 S   0.0  0.3  0:00.19 /lib/systemd/systemd-udev
 391 root         20    0   282M  27100   9072 S   0.0  1.3  0:00.00 /sbin/multipathd -d -s
 392 root         20    0   282M  27100   9072 S   0.0  1.3  0:00.00 /sbin/multipathd -d -s
 393 root         20    0   282M  27100   9072 S   0.0  1.3  0:00.00 /sbin/multipathd -d -s
 394 root         20    0   282M  27100   9072 S   0.0  1.3  0:00.05 /sbin/multipathd -d -s
 395 root         20    0   282M  27100   9072 S   0.0  1.3  0:00.00 /sbin/multipathd -d -s
 487 systemd-t    20    0 89364   6596   5788 S   0.0  0.3  0:00.09 /lib/systemd/systemd-timesyncd
 512 systemd-t    20    0 89364   6596   5788 S   0.0  0.3  0:00.00 /lib/systemd/systemd-timesyncd
 611 systemd-n    20    0 16132   8132   7132 S   0.0  0.4  0:00.05 /lib/systemd/systemd-networkd
 613 systemd-r    20    0 26340  13508   8516 S   0.0  0.7  0:00.13 /lib/systemd/systemd-resolved
 650 root         20    0   7288   2836   2600 S   0.0  0.1  0:00.01 /usr/sbin/cron -f -P
 651 messagebu    20    0   8664   4772   4164 S   0.0  0.2  0:00.10 @dbus-daemon --system --address=systemd
 669 root         20    0 82772   3900   3556 S   0.0  0.2  0:00.04 /usr/sbin/irqbalance --foreground
 671 root         20    0 33168  19104  10244 S   0.0  1.0  0:00.13 /usr/bin/python3 /usr/bin/networkd-disp
 673 root         20    0 80056   1796   1600 S   0.0  0.1  0:00.00 /usr/sbin/qemu-ga
 674 syslog       20    0   217M  5680   4596 S   0.0  0.3  0:00.03 /usr/sbin/rsyslogd -n -iNONE
 677 root         20    0 1295M  36520  24004 S   0.0  1.8  0:00.58 /usr/lib/napd/napd
 679 root         20    0 80056   1796   1600 S   0.0  0.1  0:00.00 /usr/sbin/qemu-ga
 680 root         20    0 15532   7440   6484 S   0.0  0.4  0:00.10 /lib/systemd/systemd-logind
 688 root         20    0   6220   1116   1032 S   0.0  0.1  0:00.00 /sbin/agetty -o -p -- \u --keep-baud 11
 698 root         20    0   6176   1108   1020 S   0.0  0.1  0:00.03 /sbin/agetty -o -p -- \u --noclear ttyl
 709 syslog       20    0   217M  5680   4596 S   0.0  0.3  0:00.01 /usr/sbin/rsyslogd -n -iNONE
 710 syslog       20    0   217M  5680   4596 S   0.0  0.3  0:00.00 /usr/sbin/rsyslogd -n -iNONE
 711 syslog       20    0   217M  5680   4596 S   0.0  0.3  0:00.00 /usr/sbin/rsyslogd -n -iNONE
 725 root         20    0 82772   3900   3556 S   0.0  0.2  0:00.00 /usr/sbin/irqbalance --foreground
 728 root         20    0 1295M  36520  24004 S   0.0  1.8  0:00.04 /usr/lib/napd/napd
 729 root         20    0 1295M  36520  24004 S   0.0  1.8  0:00.07 /usr/lib/napd/napd
 730 root         20    0 1295M  36520  24004 S   0.0  1.8  0:00.02 /usr/lib/napd/napd
 732 root         20    0 1295M  36520  24004 S   0.0  1.8  0:00.00 /usr/lib/napd/napd
 738 root         20    0 15436   9544   7912 S   0.0  0.5  0:00.02 sshd: /usr/sbin/sshd -D [listener] 0 of
 746 root         20    0  107M  21500  13376 S   0.0  1.1  0:00.11 /usr/bin/python3 /usr/share/unattended-
 756 root         20    0  245M  36260  28068 S   0.0  1.8  0:00.09 /usr/sbin/apache2 -k start
 760 root         20    0  229M  8288   7020 S   0.0  0.4  0:00.03 /usr/libexec/polkitd --no-debug
 766 root         20    0  229M  8288   7020 S   0.0  0.4  0:00.00 /usr/libexec/polkitd --no-debug
 770 root         20    0  229M  8288   7020 S   0.0  0.4  0:00.00 /usr/libexec/polkitd --no-debug
 779 root         20    0 1295M  36520  24004 S   0.0  1.8  0:00.00 /usr/lib/napd/napd
 794 root         20    0 1295M  36520  24004 S   0.0  1.8  0:00.10 /usr/lib/napd/napd
 804 root         20    0 1295M  36520  24004 S   0.0  1.8  0:00.11 /usr/lib/napd/napd
 814 root         20    0  107M  21500  13376 S   0.0  1.1  0:00.00 /usr/bin/python3 /usr/share/unattended-
 888 mysql      20    0 1742M  385M  36256 S   0.0 19.7  0:00.04 /usr/sbin/mysqld
 891 mysql      20    0 1742M  385M  36256 S   0.0 19.7  0:00.03 /usr/sbin/mysqld
 893 mysql      20    0 1742M  385M  36256 S   0.0 19.7  0:00.04 /usr/sbin/mysqld
 894 mysql      20    0 1742M  385M  36256 S   0.0 19.7  0:00.04 /usr/sbin/mysqld
 895 mysql      20    0 1742M  385M  36256 S   0.0 19.7  0:00.04 /usr/sbin/mysqld

1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit
```

VmStat merupakan perintah pada Linux yang berfungsi untuk menampilkan penggunaan memori, swap,

memberikan informasi interrupt system, kecepatan I/O, statistik CPU secara real time, dan proses apa saja yang sedang berjalan.

Ketikkan Command dibawah ini Terminal:

> vmstat

```
root@nayakvpc:/home/nayak# vmstat
procs -----memory----- ---swap-- -----io---- -system-- -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs  us  sy  id  wa  st
 2   0       0 991320 23800 456040    0    0   310   44  158  294   4   1  92   2   0
root@nayakvpc:/home/nayak#
```

GoAccess is an open source real-time web log analyzer and interactive viewer that runs in a terminal in unix systems or through your browser.

Install dan cek versi dari repositori Ubuntu

- sudo apt install goaccess -y
- goaccess -version
- goaccess -V

```

root@nayakvpc: /home/nayak
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Hit:4 https://ppa.launchpadcontent.net/ondrej/php/ubuntu jammy InRelease
Get:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3439 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3174 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [519 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [19.6 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1268 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [448 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadata [14.2 kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1029 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [22.8 kB]
Fetched 10.3 MB in 6s (1675 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
7 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-5.15.0-173 linux-headers-5.15.0-173-generic linux-image-5.15.0-173-generic
  linux-modules-5.15.0-173-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  goaccess
0 upgraded, 1 newly installed, 0 to remove and 7 not upgraded.
Need to get 340 kB of archives.
After this operation, 1139 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 goaccess amd64 1:1.5.5-1 [340 kB]
Fetched 340 kB in 2s (178 kB/s)
Selecting previously unselected package goaccess.
(Reading database ... 127767 files and directories currently installed.)
Preparing to unpack .../goaccess_1%3a1.5.5-1_amd64.deb ...
Unpacking goaccess (1:1.5.5-1) ...
Setting up goaccess (1:1.5.5-1) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@nayakvpc:/home/nayak# goaccess -version
goaccess: invalid option -- 'v'
root@nayakvpc:/home/nayak# goaccess -V
GoAccess - 1.5.5.
For more details visit: https://goaccess.io/
Copyright (C) 2009-2022 by Gerardo Orellana

Build configure arguments:
  --enable-utf8
  --enable-geoip=mmdb
  --with-openssl
root@nayakvpc:/home/nayak#

```

Default log Apache ada di: /var/log/apache2/access.

➤ `sudo tail -f /var/log/apache2/access.log`

```
t@nayakvpc:/home/nayak$ sudo tail -f /var/log/apache2/access.log
1230.177.142 - - [27/Apr/2026:15:14:27 +0000] "-" 408 3292 "-" "-"
101.172.159 - - [27/Apr/2026:15:15:59 +0000] "GET /.git/config HTTP/1.1" 404 493 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
101.172.159 - - [27/Apr/2026:15:16:01 +0000] "GET /.git/config HTTP/1.1" 404 3820 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
```

Jalankan GoAccess di Terminal:

➤ `sudo goaccess /var/log/apache2/access.log --log-format=COMBINED`

Format COMBINED sesuai dengan default Apache log format:

`%h %^[%d:%t %^] "%r" %s %b "%R" "%u"`

```
root@nayakvpc: /home/nayak

Dashboard - Overall Analyzed Requests (27/Apr/2026 - 27/Apr/2026) [Active Panel: Visitors]

Total Requests 3 Unique Visitors 0 Requested Files 1 Referrers 0
Valid Requests 3 Log Parsing Time 1s Static Files 0 Log Size 463.0 B
Failed Requests 0 Excl. IP Hits 0 Not Found 1 Tx. Amount 7.43 KiB
Log Source /var/log/apache2/access.log

> 1 - Unique visitors per day - Including spiders Total: 1/1

Hits h% Vis. v% Tx. Amount Data
-----
3 100.00% 0 0.00% 7.43 KiB 27/Apr/2026 |

2 - Requested Files (URLs) Total: 1/1

Hits h% Vis. v% Tx. Amount Mtd Proto Data
-----
1 33.33% 0 0.00% 3.21 KiB --- --- -

3 - Static Requests Total: 0/0

4 - Not Found URLs (404s) Total: 1/1

Hits h% Vis. v% Tx. Amount Mtd Proto Data
-----
2 66.67% 0 0.00% 4.21 KiB GET HTTP/1.1 /.git/config

5 - Visitor Hostnames and IPs Total: 2/2

Hits h% Vis. v% Tx. Amount Data
-----
2 66.67% 0 0.00% 4.21 KiB 46.101.172.159 |
[?] Help [Enter] Exp. Panel 0/r - 27/Apr/2026:15:23:58 [q]uit GoAccess 1.5.5
```

GoAccess bisa menampilkan laporan real-time dalam bentuk HTML.

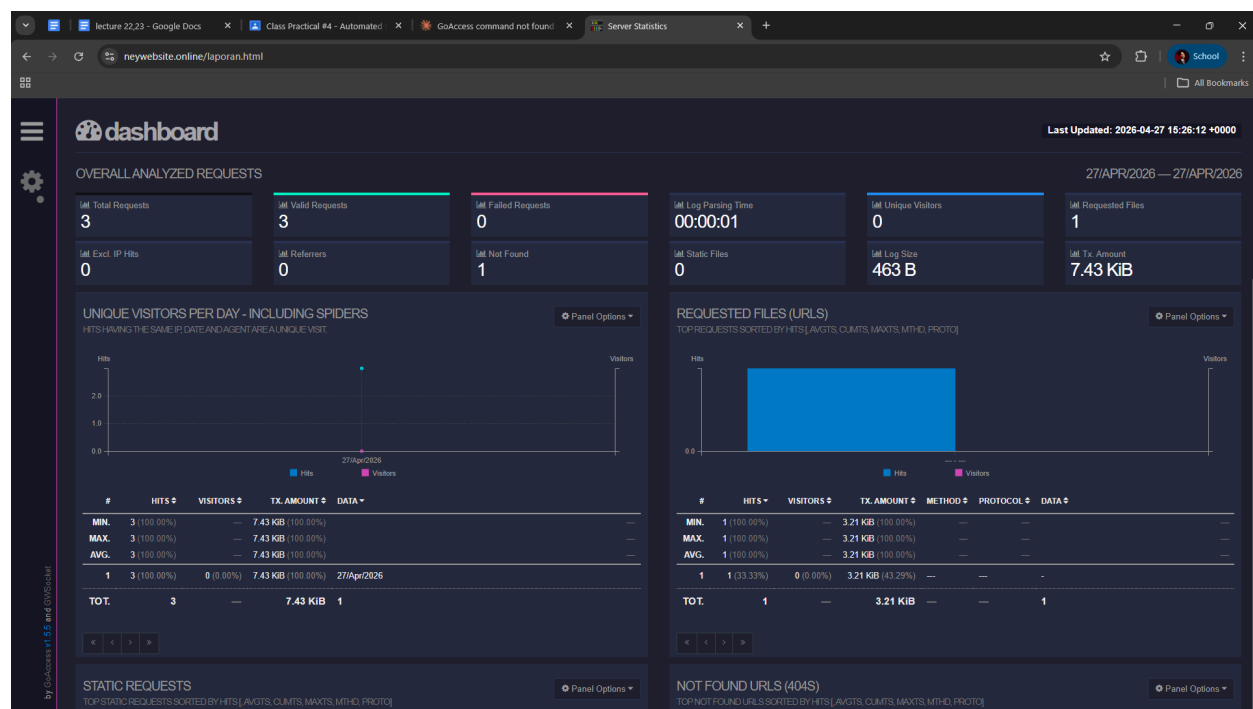
- [1] Generate report statis (HTML biasa)
 - sudo goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/html/laporan.html
- [2] Generate report real-time (pakai WebSocket server GoAccess)
 - sudo goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/html/laporan.html --real-time-html
- [3] Lalu buka di browser:

<https://neywebsite.online/laporan.html>

```
root@nayakvpc:/home/nayak# sudo goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/html/laporan.html
[PARSING /var/log/apache2/access.log] {0} @ {0/s}
Cleaning up resources...
root@nayakvpc:/home/nayak# sudo goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/html/laporan.html --real-time-html
[SETTING UP STORAGE laporan.html] {0} @ {0/s}
GoAccess - version 1.5.5 - Feb  5 2022 18:37:15
Config file: /etc/goaccess/goaccess.conf

Fatal error has occurred
Error occurred at: src/parser.c - read_log - 2015
Unable to open the specified log file 'laporan.html'. No such file or directory

root@nayakvpc:/home/nayak# sudo goaccess /var/log/apache2/access.log \
--log-format=COMBINED \
-o /var/www/html/laporan.html \
--real-time-html
[PARSING /var/log/apache2/access.log] {0} @ {0/s}
Websocket server ready to accept new client connections
C
SIGINT caught!
Closing GoAccess...
Stopping WebSocket server...
Cleaning up resources...
root@nayakvpc:/home/nayak# sudo goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/html/laporan.html --real-time-html
Websocket server ready to accept new client connections
```



[1] Buat service supaya GoAccess selalu jalan di background:

➤ sudo nano /etc/systemd/system/goaccess.service

[2] Open file dan copy text di bawah ini dan Isi dengan:

[Unit]

Description=GoAccess Real-Time Log Analyzer

After=network.target

[Service]

ExecStart=/usr/bin/goaccess /var/log/apache2/access.log --log-format=COMBINED -o
/var/www/html/report.html --real-
time-html

Restart=always

[Install]

WantedBy=multi-user.target

```
root@nayakvpc: /home/nayak
GNU nano 6.2 /etc/systemd/system/goaccess.service *
[Unit]
Description=GoAccess Real-Time Log Analyzer
After=network.target
[Service]

ExecStart=/usr/bin/goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/html/report.html
time-html

Restart=always
[Install]
WantedBy=multi-user.target

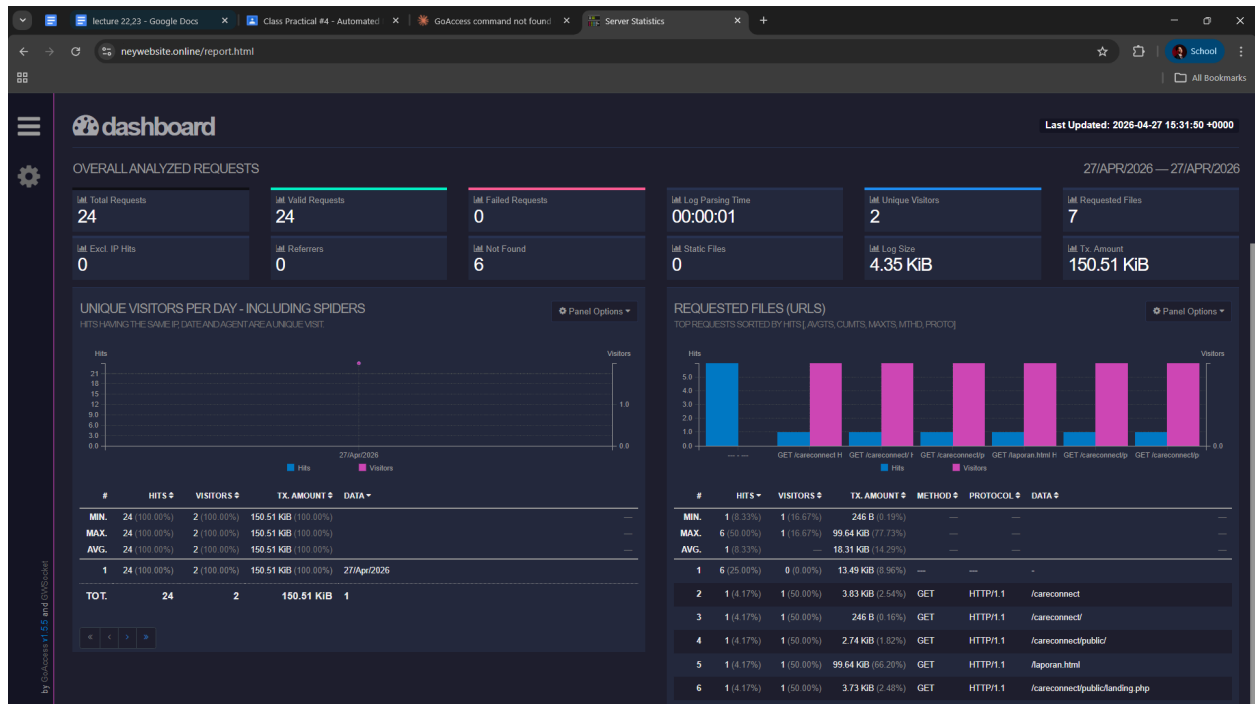
File Name to Write: /etc/systemd/system/goaccess.service
^G Help      M-D DOS Format  M-A Append     M-B Backup File
^C Cancel    M-M Mac Format  M-P Prepend    ^T Browse
```

[3] Simpan, lalu jalankan:

- sudo systemctl daemon-reload
- sudo systemctl enable goaccess
- sudo systemctl start goaccess

[4] Buka di browser:

<https://neywebsite.online/report.html>



Type Ubuntu Command :

- sudo su
- cat /var/log/apache2/error.log
- nano /var/log/apache2/error.log

```

root@nayakvpc: /home/nayak
5.0 (iPhone; CPU iPhone OS 18_6_2 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) CriOS/147.0.7
27.99 Mobile/15E148 Safari/604.1"
25.165.123.70 - - [27/Apr/2026:15:30:12 +0000] "GET /careconnect/public/landing.php HTTP/1.1" 200 3823
https://neywebsite.online/careconnect/public/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/5
7.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
03.175.188.42 - - [27/Apr/2026:15:30:19 +0000] "-" 408 3292 "-" "-"
6.85.225.42 - - [27/Apr/2026:15:30:19 +0000] "-" 408 323 "-" "-"
25.165.123.70 - - [27/Apr/2026:15:30:26 +0000] "GET /careconnect/public/patient/login.php HTTP/1.1" 200
4617 "https://neywebsite.online/careconnect/public/landing.php" "Mozilla/5.0 (Windows NT 10.0; Win64; x
4) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
25.165.123.70 - - [27/Apr/2026:15:30:46 +0000] "-" 408 323 "-" "-"
6.85.205.214 - - [27/Apr/2026:15:31:58 +0000] "GET /report.html HTTP/1.1" 200 99966 "-" "Mozilla/5.0 (W
indows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
25.165.123.70 - - [27/Apr/2026:15:32:17 +0000] "GET /careconnect/public/ HTTP/1.1" 200 9008 "-" "WhatsA
p/2.2615.101 W"
6.85.205.214 - - [27/Apr/2026:15:32:18 +0000] "-" 408 323 "-" "-"
6.85.205.208 - - [27/Apr/2026:15:32:18 +0000] "-" 408 323 "-" "-"
25.165.123.70 - - [27/Apr/2026:15:32:25 +0000] "GET /careconnect/public/ HTTP/1.1" 200 6102 "-" "Mozill
/5.0 (iPhone; CPU iPhone OS 18_6_2 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.6
Mobile/15E148 Safari/604.1"
25.165.123.70 - - [27/Apr/2026:15:32:30 +0000] "GET /favicon.ico HTTP/1.1" 404 518 "https://neywebsite.
online/careconnect/public/" "Mozilla/5.0 (iPhone; CPU iPhone OS 18_6_2 like Mac OS X) AppleWebKit/605.1.
5 (KHTML, like Gecko) Version/18.6 Mobile/15E148 Safari/604.1"
25.165.123.70 - - [27/Apr/2026:15:32:35 +0000] "GET /careconnect/public/landing.php HTTP/1.1" 200 3499
https://neywebsite.online/careconnect/public/" "Mozilla/5.0 (iPhone; CPU iPhone OS 18_6_2 like Mac OS X
AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.6 Mobile/15E148 Safari/604.1"
3.153.96.233 - - [27/Apr/2026:15:32:42 +0000] "GET / HTTP/1.1" 200 2751 "-" "Mozilla/5.0 (iPhone; CPU i
hone OS 13_2_3 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/13.0.3 Mobile/15E148 Saf
ri/604.1"
root@nayakvpc: /home/nayak# nano /var/log/apache2/access.log
GNU nano 6.2 /var/log/apache2/access.log
56.230.177.142 - - [27/Apr/2026:15:14:27 +0000] "-" 408 3292 "-" "-"
6.101.172.159 - - [27/Apr/2026:15:15:59 +0000] "GET /.git/config HTTP/1.1" 404 493 "-" "Mozilla/5.0 (X>
6.101.172.159 - - [27/Apr/2026:15:16:01 +0000] "GET /.git/config HTTP/1.1" 404 3820 "-" "Mozilla/5.0 (>
6.85.225.42 - - [27/Apr/2026:15:27:06 +0000] "GET /report.html HTTP/1.1" 404 493 "-" "Mozilla/5.0 (Win>
6.85.225.42 - - [27/Apr/2026:15:27:06 +0000] "GET /favicon.ico HTTP/1.1" 404 492 "http://103.179.56.17>
25.165.123.70 - - [27/Apr/2026:15:28:03 +0000] "GET /careconnect HTTP/1.1" 301 3919 "-" "Mozilla/5.0 (>
25.165.123.70 - - [27/Apr/2026:15:28:03 +0000] "GET /careconnect/ HTTP/1.1" 302 246 "-" "Mozilla/5.0 (>
25.165.123.70 - - [27/Apr/2026:15:28:03 +0000] "GET /careconnect/public/ HTTP/1.1" 200 2809 "-" "Mozil>
6.50.157.4 - - [27/Apr/2026:15:28:07 +0000] "GET /careconnect HTTP/1.1" 404 496 "https://www.google.co>
6.50.157.4 - - [27/Apr/2026:15:28:07 +0000] "GET /favicon.ico HTTP/1.1" 404 495 "http://neywebsite.onl>
25.165.123.70 - - [27/Apr/2026:15:28:10 +0000] "GET /favicon.ico HTTP/1.1" 404 842 "https://neywebsite>
25.165.123.70 - - [27/Apr/2026:15:28:30 +0000] "-" 408 3292 "-" "-"
25.165.123.70 - - [27/Apr/2026:15:28:30 +0000] "-" 408 3292 "-" "-"
6.85.205.208 - - [27/Apr/2026:15:28:32 +0000] "GET /report.html HTTP/1.1" 404 493 "-" "Mozilla/5.0 (Wi>
25.165.123.70 - - [27/Apr/2026:15:29:26 +0000] "GET /careconnet HTTP/1.1" 404 3811 "-" "Mozilla/5.0 (i>
25.165.123.70 - - [27/Apr/2026:15:29:26 +0000] "GET /favicon.ico HTTP/1.1" 404 3811 "-" "Mozilla/5.0 (>
25.165.123.70 - - [27/Apr/2026:15:29:37 +0000] "GET /careconnet HTTP/1.1" 404 3811 "-" "Mozilla/5.0 (i>
6.85.205.214 - - [27/Apr/2026:15:29:59 +0000] "GET /laporan.html HTTP/1.1" 200 102030 "-" "Mozilla/5.0>
25.165.123.70 - - [27/Apr/2026:15:29:59 +0000] "GET /careconnet/public HTTP/1.1" 404 3811 "-" "Mozill>
25.165.123.70 - - [27/Apr/2026:15:30:12 +0000] "GET /careconnect/public/landing.php HTTP/1.1" 200 3823>
03.175.188.42 - - [27/Apr/2026:15:30:19 +0000] "-" 408 3292 "-" "-"
6.85.225.42 - - [27/Apr/2026:15:30:19 +0000] "-" 408 323 "-" "-"
25.165.123.70 - - [27/Apr/2026:15:30:26 +0000] "GET /careconnect/public/patient/login.php HTTP/1.1" 20>
25.165.123.70 - - [27/Apr/2026:15:30:46 +0000] "-" 408 323 "-" "-"
6.85.205.214 - - [27/Apr/2026:15:31:58 +0000] "GET /report.html HTTP/1.1" 200 99966 "-" "Mozilla/5.0 (>
25.165.123.70 - - [27/Apr/2026:15:32:17 +0000] "GET /careconnect/public/ HTTP/1.1" 200 9008 "-" "Whats>
6.85.205.214 - - [27/Apr/2026:15:32:18 +0000] "-" 408 323 "-" "-"
6.85.205.208 - - [27/Apr/2026:15:32:18 +0000] "-" 408 323 "-" "-"
25.165.123.70 - - [27/Apr/2026:15:32:25 +0000] "GET /careconnect/public/ HTTP/1.1" 200 6102 "-" "Mozil>
25.165.123.70 - - [27/Apr/2026:15:32:30 +0000] "GET /favicon.ico HTTP/1.1" 404 518 "https://neywebsite>

```

Monitor file error.log Apache secara real-time:

➤ `tail -f /var/log/apache2/error.log`

Lakukan monitoring pada Server:

- `sudo su`
- `sudo cat /var/log/auth.log | grep "denied"`
- `sudo cat /var/log/auth.log | grep "192.3.228.147" # IP address tersangka`
- `sudo cat /var/log/auth.log | grep "Accepted publickey"`
- `sudo cat /var/log/auth.log | grep "Connection closed by authenticating"`
- `sudo cat /var/log/auth.log | grep "Connection closed by invalid user"`
- `sudo zgrep "Failed password" /var/log/auth.log*`
- `sudo zgrep "204.76.203.28" /var/log/auth.log*`
- `sudo zgrep -E "Failed password for (admin|root|sql)" /var/log/auth.log*`
- `sudo zgrep -E "Received disconnect from" /var/log/auth.log*`

```

root@nayakvpc: /home/nayak
125.165.123.70 - - [27/Apr/2026:15:32:17 +0000] "GET /careconnect/public/ HTTP/1.1" 200 9008 "-" "WhatsApp/2.2615.101 W"
36.85.205.214 - - [27/Apr/2026:15:32:18 +0000] "-" 408 323 "-" "-"
36.85.205.208 - - [27/Apr/2026:15:32:18 +0000] "-" 408 323 "-" "-"
125.165.123.70 - - [27/Apr/2026:15:32:25 +0000] "GET /careconnect/public/ HTTP/1.1" 200 6102 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 18_6_2 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.6 Mobile/15E148 Safari/604.1"
125.165.123.70 - - [27/Apr/2026:15:32:30 +0000] "GET /favicon.ico HTTP/1.1" 404 518 "https://neywebsite.online/careconnect/public/" "Mozilla/5.0 (iPhone; CPU iPhone OS 18_6_2 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.6 Mobile/15E148 Safari/604.1"
125.165.123.70 - - [27/Apr/2026:15:32:35 +0000] "GET /careconnect/public/landing.php HTTP/1.1" 200 3499 "https://neywebsite.online/careconnect/public/" "Mozilla/5.0 (iPhone; CPU iPhone OS 18_6_2 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.6 Mobile/15E148 Safari/604.1"
43.153.96.233 - - [27/Apr/2026:15:32:42 +0000] "GET / HTTP/1.1" 200 2751 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 13_2_3 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/13.0.3 Mobile/15E148 Safari/604.1"
root@nayakvpc:/home/nayak# tail -f /var/log/apache2/error.log
[Mon Apr 27 15:08:28.870170 2026] [mpm_prefork:notice] [pid 756] AH00163: Apache/2.4.52 (Ubuntu) OpenSSL/3.0.2 configured -- resuming normal operations
[Mon Apr 27 15:08:28.881560 2026] [core:notice] [pid 756] AH00094: Command line: '/usr/sbin/apache2'
^C
root@nayakvpc:/home/nayak# sudo cat /var/log/auth.log | grep "denied"
> ^C
root@nayakvpc:/home/nayak# sudo cat /var/log/auth.log | grep "Accepted publickey"
root@nayakvpc:/home/nayak# sudo cat /var/log/auth.log | grep "Connection closed by authenticating"
Apr 23 05:33:15 nayakvpc sshd[1013]: Connection closed by authenticating user nayak 103.175.188.42 port 64082 [preauth]
root@nayakvpc:/home/nayak# sudo cat /var/log/auth.log | grep "Connection closed by invalid user"
Apr 23 00:08:50 nayakvpc sshd[26481]: Connection closed by invalid user admin 185.246.130.20 port 23036 [preauth]
root@nayakvpc:/home/nayak# sudo zgrep "Failed password" /var/log/auth.log*
Apr 22 23:42:10 nayakvpc sshd[1145]: Failed password for nayak from 156.230.177.142 port 56566 ssh2
Apr 23 00:07:15 nayakvpc sshd[26473]: Failed password for invalid user 0 from 185.246.130.20 port 13288 ssh2
Apr 23 00:08:42 nayakvpc sshd[26481]: Failed password for invalid user admin from 185.246.130.20 port 23036 ssh2
Apr 23 00:43:34 nayakvpc sshd[32287]: Failed password for nayak from 36.85.205.208 port 56449 ssh2
Apr 23 00:44:28 nayakvpc sshd[32287]: Failed password for nayak from 36.85.205.208 port 56449 ssh2
Apr 23 00:50:53 nayakvpc sshd[32366]: Failed password for nayak from 103.175.188.42 port 51776 ssh2
Apr 23 00:51:04 nayakvpc sshd[32366]: Failed password for nayak from 103.175.188.42 port 51776 ssh2
Apr 23 01:26:19 nayakvpc sshd[32672]: Failed password for nayak from 36.85.225.129 port 54886 ssh2
Apr 23 03:56:02 nayakvpc sshd[33437]: Failed password for nayak from 36.85.225.129 port 61367 ssh2
Apr 23 04:34:09 nayakvpc sshd[33735]: Failed password for nayak from 36.85.205.214 port 63268 ssh2
Apr 23 05:32:25 nayakvpc sshd[1013]: Failed password for nayak from 103.175.188.42 port 64082 ssh2
Apr 23 05:32:56 nayakvpc sshd[1013]: message repeated 2 times: [ Failed password for nayak from 103.175.188.42 port 64082 ssh2]
Apr 23 05:33:02 nayakvpc sshd[1013]: Failed password for nayak from 103.175.188.42 port 64082 ssh2
Apr 23 05:33:13 nayakvpc sshd[1013]: Failed password for nayak from 103.175.188.42 port 64082 ssh2
Apr 23 05:33:44 nayakvpc sshd[1018]: Failed password for nayak from 103.175.188.42 port 63363 ssh2
Apr 27 15:16:03 nayakvpc sshd[1322]: Failed password for nayak from 36.85.205.214 port 58969 ssh2
Apr 27 15:18:22 nayakvpc sshd[1401]: Failed password for nayak from 36.85.225.42 port 65055 ssh2
Apr 27 15:35:54 nayakvpc sudo: root : TTY=pts/1 ; PWD=/home/nayak ; USER=root ; COMMAND=/usr/bin/zgrep 'Failed password' /var/log/auth.log
root@nayakvpc:/home/nayak# sudo zgrep "204.76.203.28" /var/log/auth.log*
Apr 27 15:36:01 nayakvpc sudo: root : TTY=pts/1 ; PWD=/home/nayak ; USER=root ; COMMAND=/usr/bin/zgrep 204.76.203.28 /var/log/auth.log
root@nayakvpc:/home/nayak# sudo zgrep -E "Failed password for (admin|root|sql)" /var/log/auth.log*
root@nayakvpc:/home/nayak# sudo zgrep -E "Received disconnect from" /var/log/auth.log*
Apr 27 15:36:10 nayakvpc sudo: root : TTY=pts/1 ; PWD=/home/nayak ; USER=root ; COMMAND=/usr/bin/zgrep -E 'Received disconnect from' /var/log/auth.log
root@nayakvpc:/home/nayak# █

```

Lecture 23

Buka

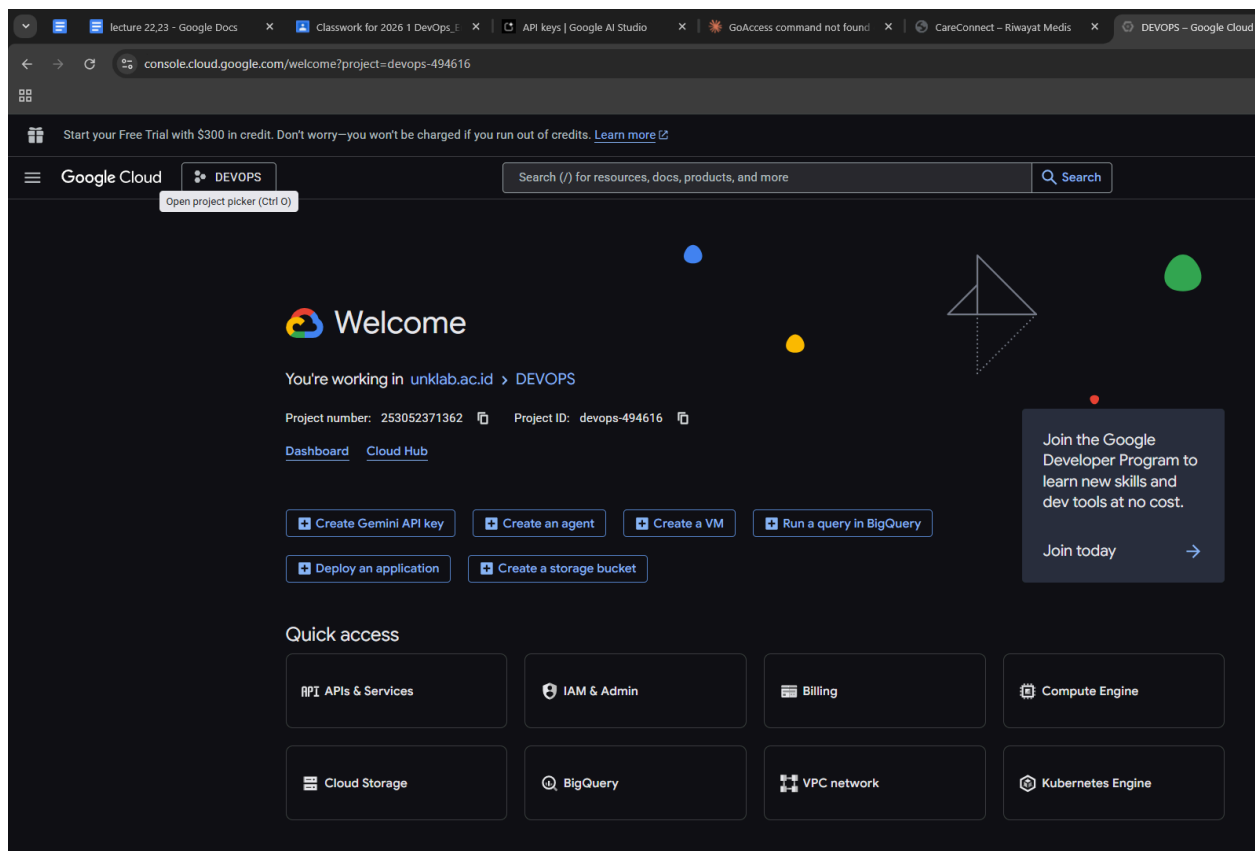
<https://console.cloud.google.com>

Jangan lupa untuk login

1. Klik pada bagian sebelah tulisan
GoogleCloud

2. Lalu Klik New Project

3. Lalu isi project namanya (misal:
DevOps). Lalu tekan tombol
Create.



Masuk ke

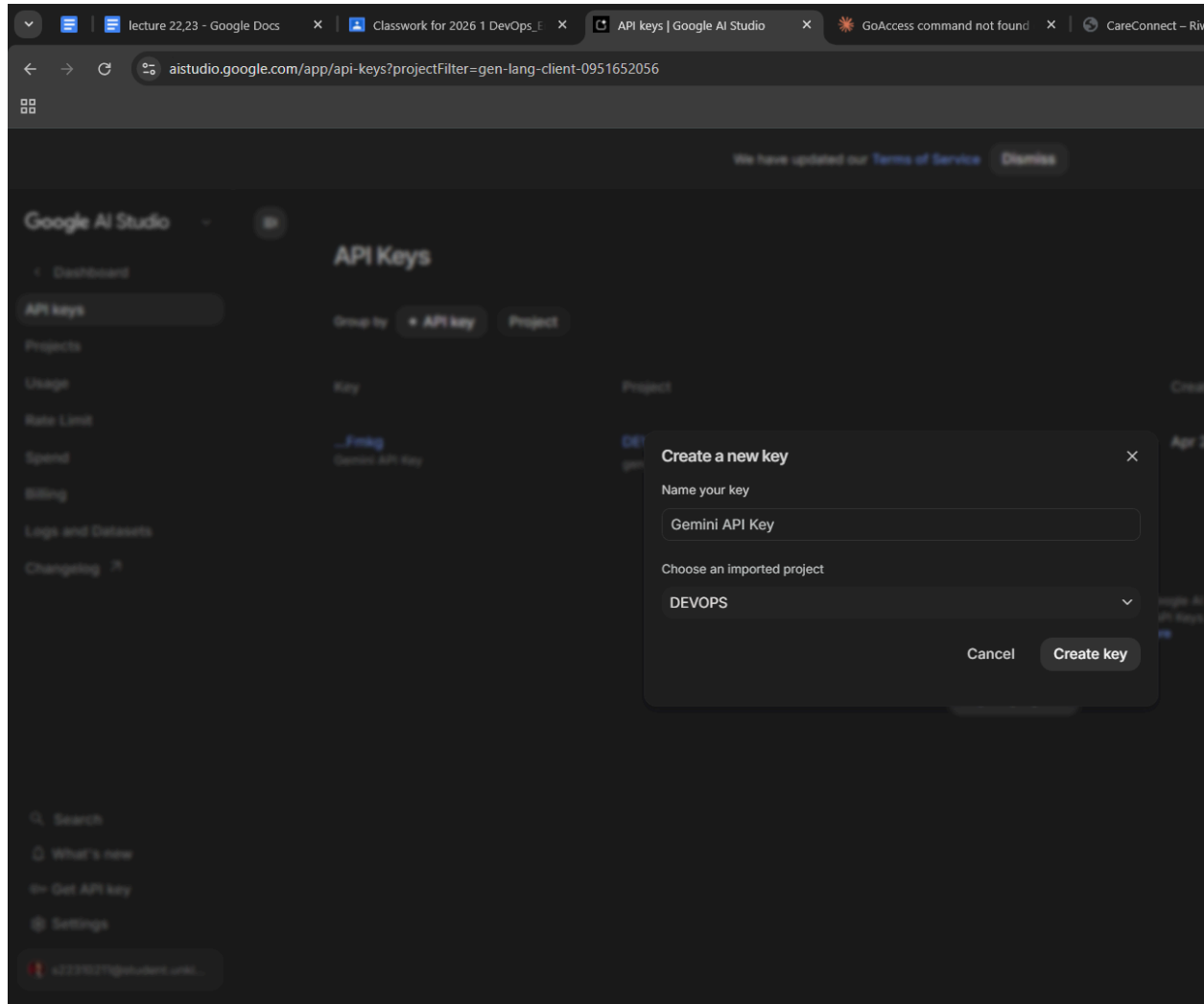
<https://makersuite.google.com/app/apikey>

2. Klik "Create API Key":

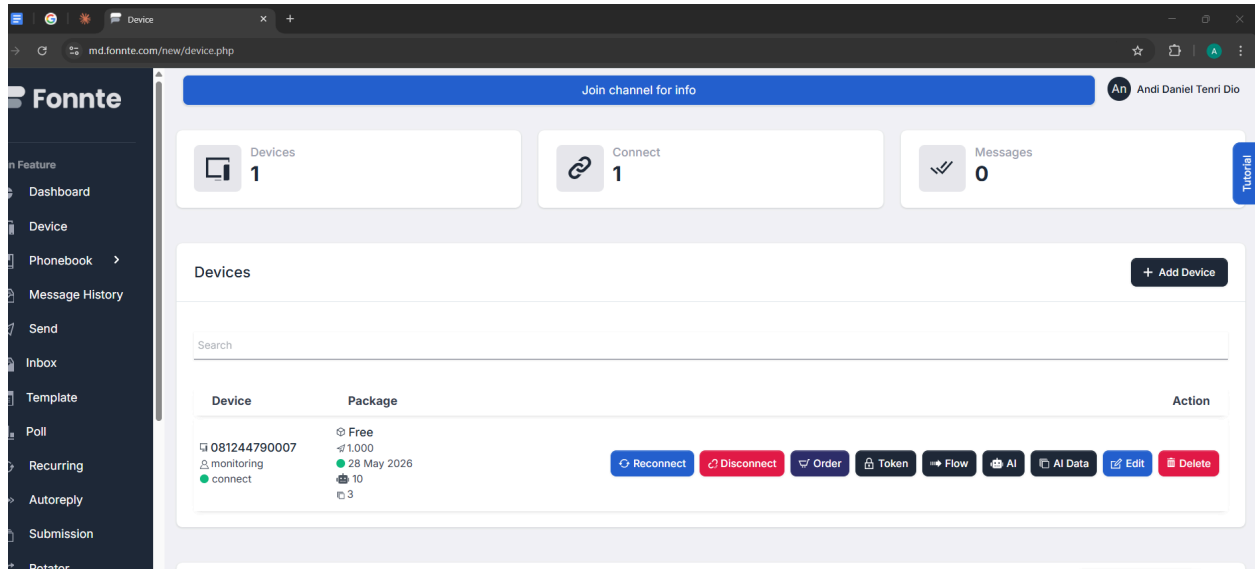
1. Lalu pilih project DevOps yang
telah kita buat sebelumnya

2. Lalu klik Create API key in
Existing Project

3. Salin dan simpan baik-baik



1. Kunjungi Website Fonnte. <https://fonnte.com>
2. Login / Daftar Akun
 - a. Kalau belum punya, daftar dulu.
 - b. Setelah login, kamu bakal masuk ke Dashboard.
3. Masuk ke Halaman Token API
 - a. Di Dashboard → Klik menu "Device"
 - b. Tekan tombol Add Device, lalu masukkan nama device dan nomor device
 - c. Lalu tekan + Add Device
4. Buat Token Baru
 - a. Pertama Connectkan dulu devicenya
 - b. Klik "+ Add Token" maka langsung akan tercopy tokennya
5. Simpan Tokennya untuk di masukkan dalam monitor.py

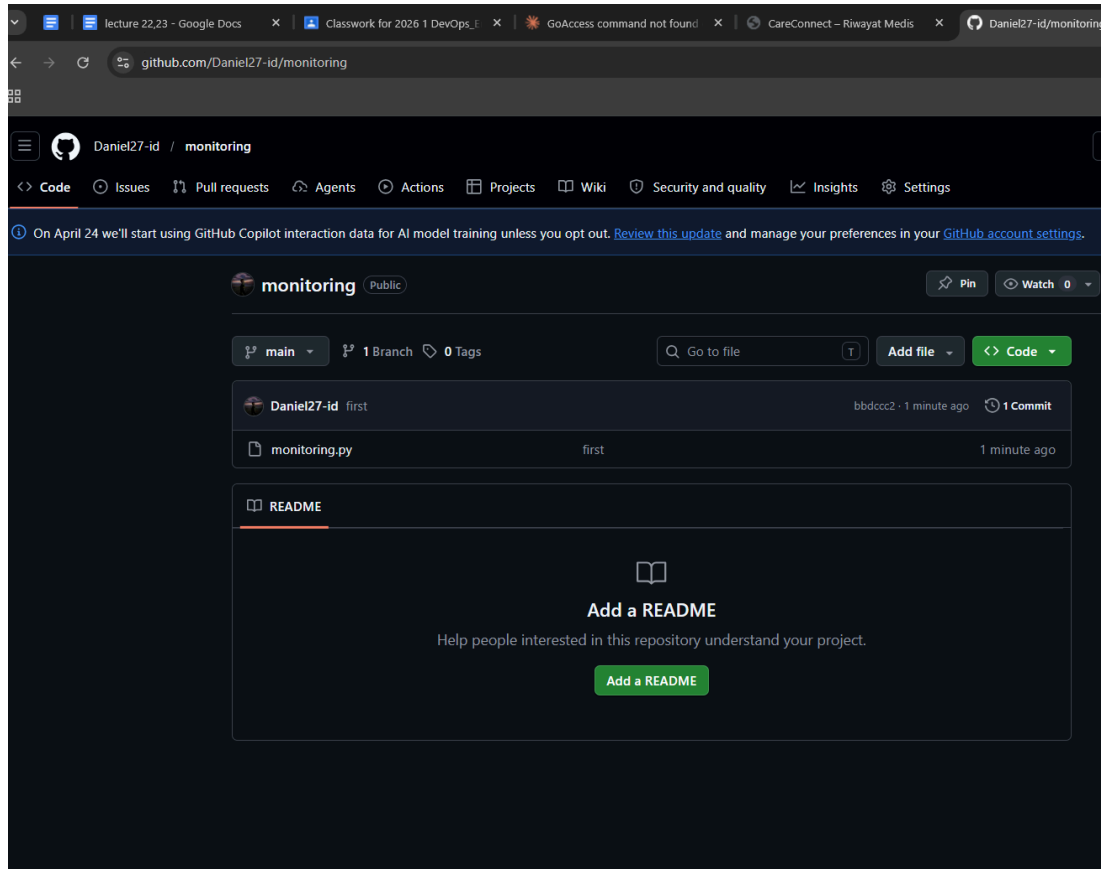


1. Masuk buat file `monitor.py` dalam local pc kita
code programnya seperti di samping

2. Isi pada bagian KonfigurasiAPI Gemini. API yang
telah kalian buat tadi.

3. Lalu pada `def send_whatsapp(message)`.
Masukkan token dari fonnte yang telah kita ambil
tadi, dan masukkan nomor device yang telah kita
hubungkan tadi.

1. Buat repo baru lalu push [monitrong.py](https://github.com/monitrong.py) ke repo yang kita buat



CONFIGURING JENKIN

Jalankan Jenkins dengan

```
sudo java -jar jenkins.war --httpPort=9090 &
```

Login ke dalam jenkinsnya

Klik new Item, lalu enter namanya. Sebebas kalian mau beri nama apa, jangan lupa select free style project

Lalu pada bagian Source Code Management:

1. Isi ssh-key dari repository yang kita buat tadi
2. Credentialnya juga
3. Lalu ganti type branch dari **master** menjadi **main**

CONFIGURING JENKIN

Lalu pada bagian triggers tekan **build periodically**

1. buat schedule untuk untuk berjalan setiap 1 hari sekali
`H/24 * * * *`

Lalu kita buat build stepnya klik **add build step** lalu pilih **execute shell**

Masukkan command seperti di ini

```
cd /root
```

```
rm -rf monitoring
```

```
git clone https://github.com/Daniel27-id/monitoring.git
```

```
cd monitoring
```

```
python3 monitoring.py
```



Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

☐ None

☒ Git ?

Repositories ?

Repository URL ?

git@github.com:Daniel27-id/monitoring.git

Credentials ?

credentials

+ Add

Advanced

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Triggers

Save

Apply

monitoring Config - Jenkins

103.179.56.171:9090/job/monitoring/configure

School

All Bookmarks

 Jenkins

monitoring

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

☐ Trigger builds remotely (e.g., from scripts) ?

☐ Build after other projects are built ?

☒ Build periodically ?

☐ Poll SCM ?

Schedule ?

H/24 * * * *

Would last have run at Tuesday, April 28, 2026, 4:13:00 AM Coordinated Universal Time; would next run at Tuesday, April 28, 2026, 4:37:00 AM Coordinated Universal Time.

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

☐ Use secret text(s) or file(s) ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

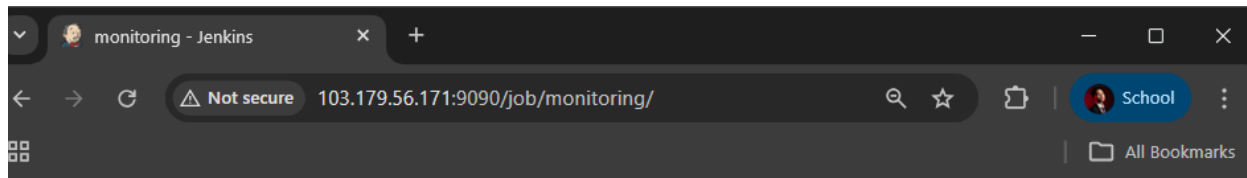
Execute shell ?

Command

See the list of available environment variables

cd /root
rm -rf monitoring
git clone https://github.com/Daniel27-id/monitoring.git
cd monitoring
python3 monitoring.py

SaveApply



 **Jenkins** / monitoring

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds >

Filter

Today

✓

#2

04.33

▼

✓

#1

04.31

▼

✓

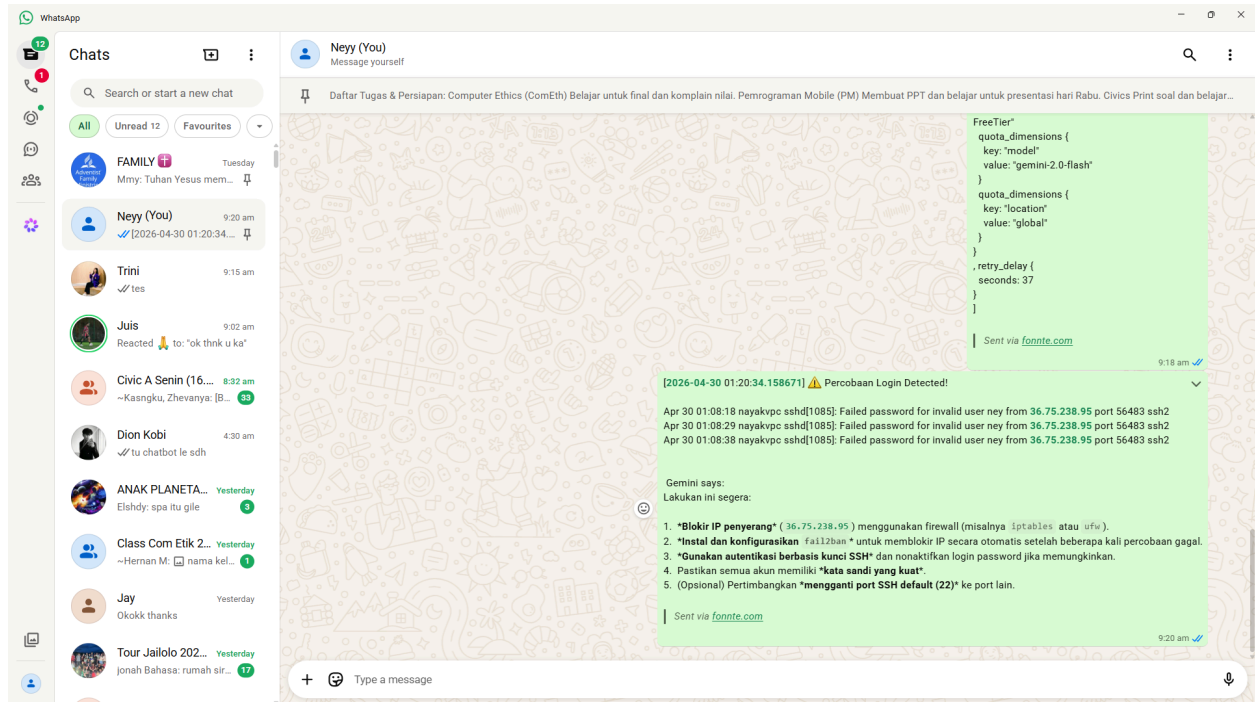
monitoring

Add description

Permalinks

- Last build (#2), 2 min 35 sec ago
- Last stable build (#2), 2 min 35 sec ago
- Last successful build (#2), 2 min 35 sec ago
- Last completed build (#2), 2 min 35 sec ago

REST API Jenkins 2.555.1



Ini outputnya

[2026-04-30 01:20:34.158671] ⚠ Percobaan Login Detected!

Apr 30 01:08:18 nayakvpc sshd[1085]: Failed password for invalid user ney from 36.75.238.95 port 56483 ssh2

Apr 30 01:08:29 nayakvpc sshd[1085]: Failed password for invalid user ney from 36.75.238.95 port 56483 ssh2

Apr 30 01:08:38 nayakvpc sshd[1085]: Failed password for invalid user ney from 36.75.238.95 port 56483 ssh2

Gemini says:

Lakukan ini segera:

1. **Blokir IP penyerang** (`36.75.238.95`) menggunakan firewall (misalnya `iptables` atau `ufw`).
2. **Instal dan konfigurasi** `fail2ban` untuk memblokir IP secara otomatis setelah beberapa kali percobaan gagal.
3. **Gunakan autentikasi berbasis kunci SSH** dan nonaktifkan login password jika memungkinkan.
4. Pastikan semua akun memiliki **kata sandi yang kuat**.
5. (Opsional) Pertimbangkan **mengganti port SSH default (22)** ke port lain.

> _Sent via fonte.com_